Sample Paper: Retrieval Quality in Long-Context RAG

Abstract. This synthetic fixture studies how chunk size affects retrieval quality when context grows long. We build a controlled corpus and measure recall at fixed budgets. The corpus contains twelve topical clusters with gradual vocabulary drift. Each cluster holds thirty synthetic paragraphs with unique markers.

1. Introduction. Retrieval augmented generation grounds model output in external evidence. Long documents challenge fixed budget retrievers because relevant evidence spreads thinly. Prior work reports degraded recall as document length grows. We hypothesize that smaller chunks sharpen boundary precision, while larger chunks preserve local cohesion. Both effects matter for answer support rate in evidence question answering.

2. Method. We index every paragraph with a fixed encoder and query each cluster with paraphrased questions. Retrieval uses dense cosine similarity over normalized vectors. We sweep chunk sizes from one hundred to two thousand characters with a fixed overlap ratio of twelve percent. Metrics include recall at five, boundary precision, and duplicate rate.

2.1 Corpus construction. Every paragraph embeds six marker tokens that never appear outside its cluster. Markers allow exact attribution of retrieved chunks without human labels.

2.2 Controls. Encoding batch order is fixed. Random seeds are pinned for all encoders. The evaluation harness replays cached model outputs to keep runs deterministic.

3. Results. Recall at five peaks near medium chunk sizes and declines at both extremes. Small chunks fragment evidence and raise duplicate retrieval. Large chunks dilute similarity and pull in irrelevant neighbors. Boundary precision improves monotonically with smaller chunks. Answer support rate follows recall closely across all settings.

4. Discussion. The tradeoff suggests adaptive strategies that respect section boundaries. Limitations. Our corpus is synthetic, so conclusions may not transfer to scanned scientific papers.

5. Conclusion. Chunk size is a first-order factor for long context retrieval quality and must be reported in evaluations.